

Meteorix

OVERVIEW

YOLO, DETR, Transformer

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YOLO

Goal : Find and locate objects in an image.

Advantages

- Super fast
- Real time
- Accurate for well-separated objects

Limitations

- Struggles with small or overlapping objects

Main idea

Before YOLO :

- scanning small parts + many analysis

YOLO :

- based on CNNs + designed to recognize patterns
- looks at the whole image once and predicts both :
 - What objects are present
 - Where they are

TRANSFORMERS

Goal : Understand **sequences of information****, like a text, a speech, or even image patches.**

Main idea

Advantages

- Simpler design and easy to implement.
- Understands the global context of the image.
- Uses attention to focus on the important parts of the input.
- Looks at the whole sequence at once
- Splits the image into many small patches
- Learns how those patches relate to each other

DETR

Goal : Detect objects in images, but in a more modern and elegant way.

Advantages

- Super fast
- Real time
- Accurate for well-separated objects

Limitations

- Takes longer to train, it needs more computing power.
- Struggles with ****small objects****.

Main idea

- Combines a CNN to get features with a Transformer to understand global context
- Looks at the whole image and learns how different sub-parts relate to each other
- Uses a special set-based matching system to make sure each object is detected only once